

Pathogen genome analysis with PathogenWatch

Version: April 2025



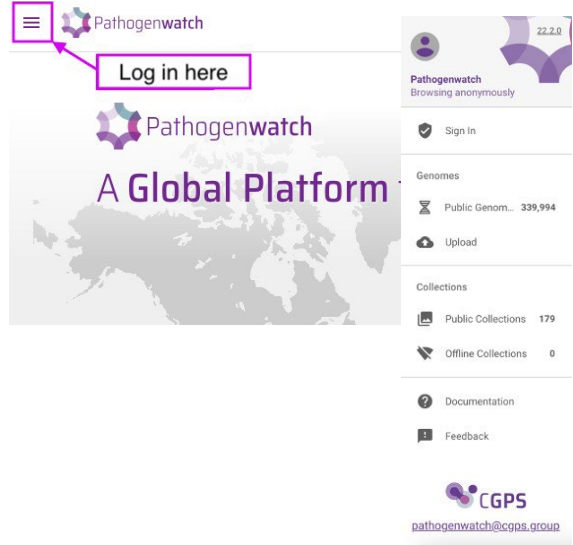
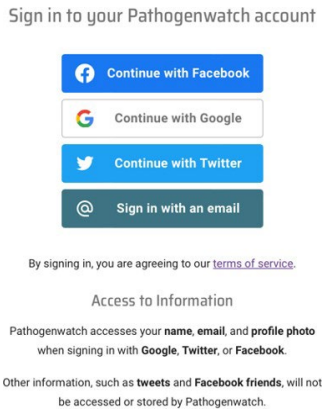
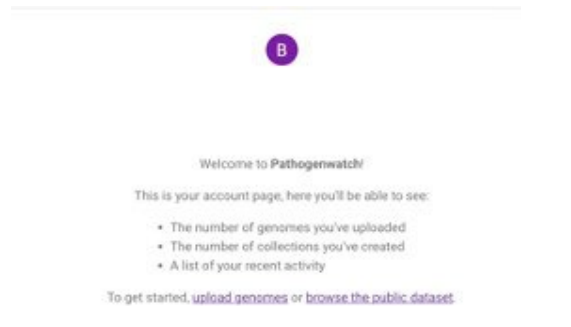
**Utrecht
University**

Introduction

Pathogenwatch (<https://pathogen.watch/>) is a web-based tool for genomic surveillance that enables users to upload and analyse their own genome data. Pathogenwatch performs a series of species-specific analytics including MLST, identification of resistance and virulence loci, replicon typing and genome clustering. MLST Sequence typing is available for over 100 species using schemes from PubMLST, Pasteur, and Enterobase. Users can create interactive visualisations of phylogenetic trees made with genome collections of publicly available genomes and including their own uploaded genomes data.

Manual to upload genomic data to Pathogenwatch

Create an PathogenWatch account

Open PathogenWatch on your browser https://pathogen.watch/ Click on “Sign In” by clicking on the icon in the upper left corner of the page. Note: it is possible to set your metadata private, see https://cgps.gitbook.io/pathogenwatch/how-to-use-pathogenwatch/private-metadata	
Create your own login credentials Check your SPAM folder if you choose to sign in by email and have difficulties receiving the login link.	
View My Account page Now that you are logged in, you will see your Account page where you can manage your genomes, collections and account settings.	

Navigating PathogenWatch

Navigate the site

- Click the menu icon

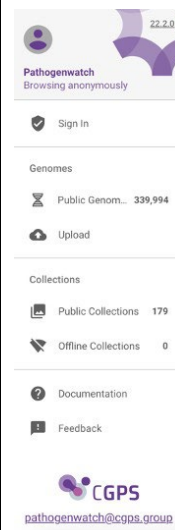
The main menu bar can be accessed by clicking on the menu icon on the upper left corner of the page.



The menu allows you to navigate the site. There are four key pages:

- Genomes
- Collections
- Upload
- Documentation

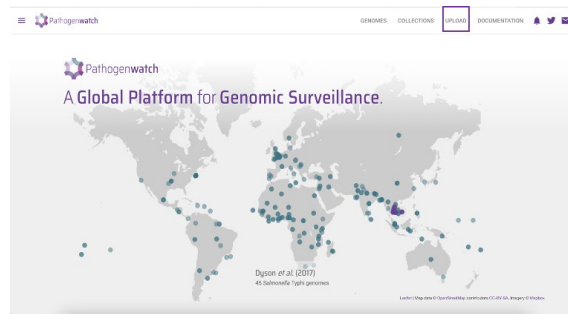
More information about navigating the site:
<https://cgps.gitbook.io/pathogenwatch>



Importing data

Navigate to the “upload” page:

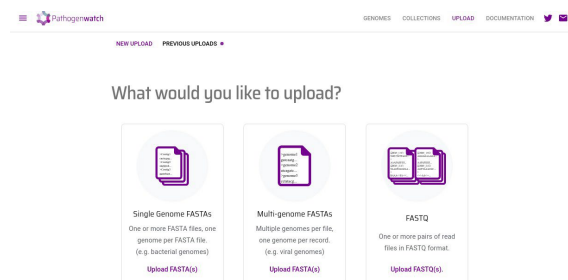
On the home screen, click on “Upload” in the upper right corner.



Select the file type to upload

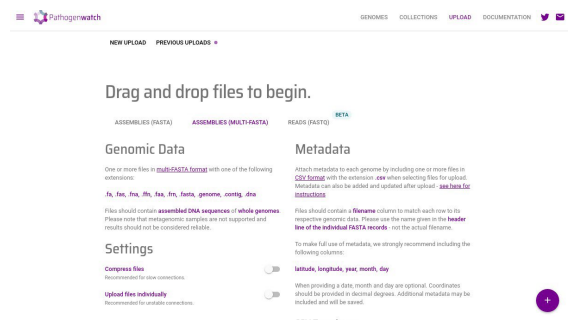
Click the “Upload” button

Click on “Single Genome FASTAs”



Drag and drop files to begin

Click the “+” button to select files to upload

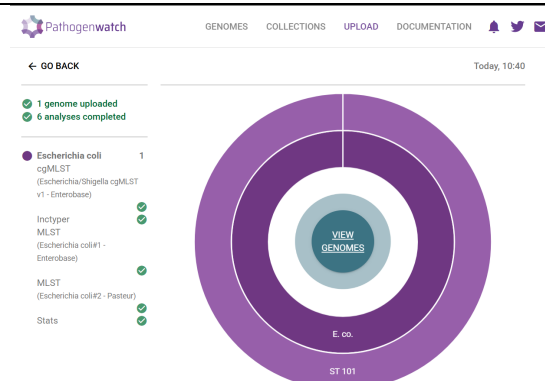


Processing screen and viewing results

Depending on the species of the genome, Pathogenwatch will give the following results:

- Species
- AMR (not for every species)
- Core summary (not every species)
- cgMLST
- Inctyper (not for every species)
- MLST

If you click the "View Genomes" button, you will be directed to the results page of all your uploaded genomes.



Viewing genomes

View your uploaded genomes

The uploaded assemblies are presented in the “view Genomes” page

Name ID	Organism	Type	Typing Schema	Country	Date ID	Access
E1_223905922-1	Staphylococcus aureus	398	MLST			Private
E_faecalis_2003_ens_1.as	Enterococcus faecalis	*24cf	MLST			Private
E.coli_3_R_assembly	Escherichia coli	101 / *430	MLST / MLST			Private

You can click the names of the uploaded genomes. This will show you the “genome report”, a detailed report of PathogenWatch analysis of the assembled genome.

DAJMS101
Escherichia coli

MLST - Multilocus sequence typing
Source: Escherichia coli#1 - Enterobase

Sequence type: **12823**
View all ST.12823

Profile: adk fumC gyrB lcd mdh purA recA
10 27 5 8 8 7 185

Alternative MLST
Source: Escherichia coli#2 - Pasteur

Sequence type: **486**
View all ST.486

Profile: dinB lcdA pabB polB putP trpA trpB uidA
10 2 3 3 166 1 4 2

Plasmid Inc types
Database sourced from <https://cge.cbs.dtu.dk/services/PlasmidFinder/>

Creating a tree with your genome and publicly available genomes

Note: only a selection of species can be used to build trees. Unfortunately, it is not possible to build a tree with *E. coli*.

In this example, we will use *Staphylococcus aureus*

Go to the page with the list of your uploaded genome(s) and select the genome that you want to include in the tree

Select only genomes of similar species/MLST STs.
For different species/STs; build different trees

Name ID	Organism	Type	Typing Schema	Country	Date ID	Access
DAJMS101	Escherichia coli	12823 / 48	MLST / MLST			Private
E.coli_206_US_assembly	Escherichia coli	*4112 / *6	MLST / MLST			Private
E.coli_2_R_assembly	Escherichia coli	*886 / *9	MLST / MLST			Private
E1_223905922-1	Staphylococcus aureus	398	MLST			Private
E1_223905922-1	Staphylococcus aureus	398	MLST			Private
E_faecalis_2003_ens_1.as	Enterococcus faecalis	*24cf	MLST			Private
E.coli_3_R_assembly	Escherichia coli	101 / *430	MLST / MLST			Private
GCA_005467206.1_ASM2	Klebsiella pneumoniae	199	MLST	US - United State	30th July 2	Public
GCA_005467206.1_ASM2	Klebsiella pneumoniae	101	MLST	TR - Turkey	28th May 2	Public
GCA_005467203.1_ASM2	Klebsiella pneumoniae	3705	MLST	US - United State	20th Janua	Public
GCA_019393005.1_ASM1	Klebsiella pneumoniae	336	MLST	US - United State	14th March	Public

1. Click on “CLEAR FILTERS” at the bottom left corner.

2. Filter the publicly available genomes of Pathogenwatch from the left sidebar. Filter the genomes with the analysis results of your genome. Next is an example:

- Genus – Staphylococcus
- Species – Staphylococcus aureus
- MLST – ST 398

3. Click on the tick that selects all genomes and then click on “Selected Genomes”.

4. You can see that the number of genomes has now increased.

The screenshot shows the Pathogenwatch interface. On the left sidebar, under 'Linked collections', 'staph test Linda' is selected. Below it, 'Staphylococcus aureus' and 'MLST: ST 398' are also selected. In the main table, the 'Name' column has a tick mark in the first row. In the top right corner, the 'Selected Genomes' button is highlighted. In the bottom left corner, the 'CLEAR FILTERS' button is highlighted. The table shows a list of Staphylococcus aureus genomes with columns for Name, Organism, Type, Typing Schema, Country, Date, and Access.

Create a collection with your genome(s) and the publicly available selected genomes by clicking on “Selected genomes” in the upper right corner and select “Create collection”

The screenshot shows the Pathogenwatch interface. In the top right corner, the 'Selected Genomes' button is highlighted. In the bottom right corner, the 'Create Collection' button is highlighted. The table shows a list of Staphylococcus aureus genomes with columns for Name, Organism, Type, Typing Schema, Country, Date, and Access.

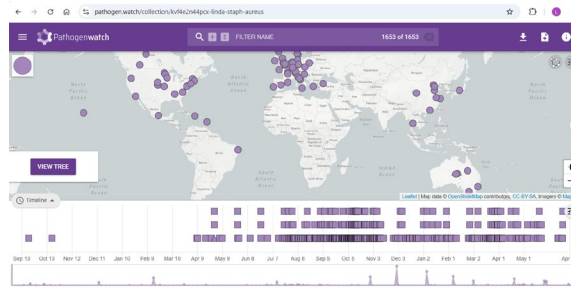
Name your collection as follows:

- Title
- Description - A short description of the collection, perhaps to explain to other people what it is for.
- PMID/DOI (optional) The [PubMed](#) or [DOI](#) identifier if it is linked to a publication.

Click "Create Now" and you will be taken to the “Collection View”.

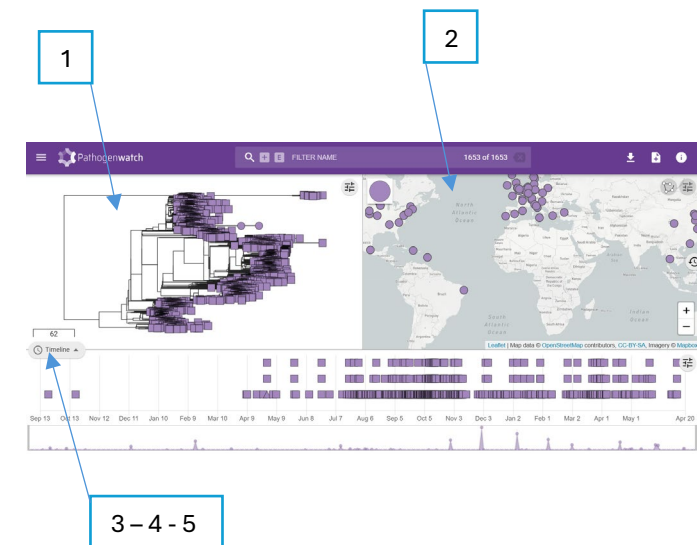
The screenshot shows the Pathogenwatch interface. In the top right corner, the 'Selected Genomes' button is highlighted. In the bottom right corner, the 'Create Collection' button is highlighted. The form shows fields for Title, Description, and PMID/DOI. The table shows a list of Staphylococcus aureus genomes with columns for Name, Organism, Type, Typing Schema, Country, Date, and Access.

If you click “view tree”, a tree will be calculated using a neighbour-joining method



Locate the five interactive components:

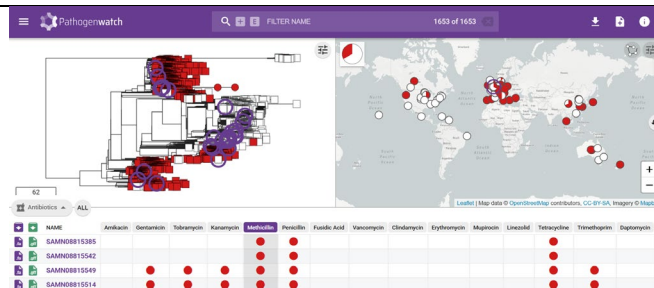
1. A phylogenetic tree.
2. A global map showing the locations of genomes that have geospatial metadata.
3. A metadata display area that includes the user-supplied data, calculated typing assignments such as MLST, and antimicrobial resistance predictions.
4. A query bar that allows the selection or finding of assemblies by searching the metadata fields.
5. A timeline showing the collection dates of assemblies that have temporal metadata.



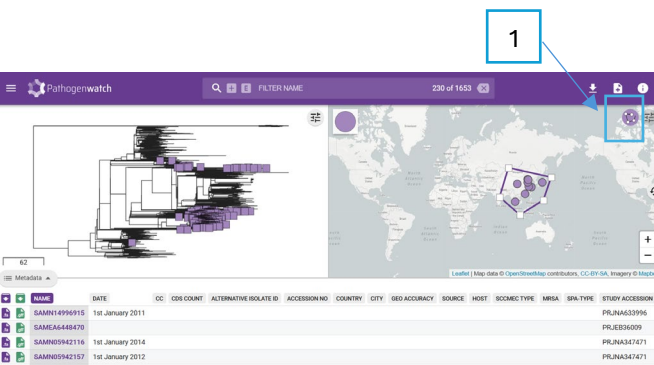
In the “Antibiotics” window, you can select a specific antibiotic.

Circles in the map will be colored present/absent (red/white) genotype or resistant/sensitive (red/white) phenotype.

The circle in the top left shows the overall ratio for the selected resistance phenotype or genotype



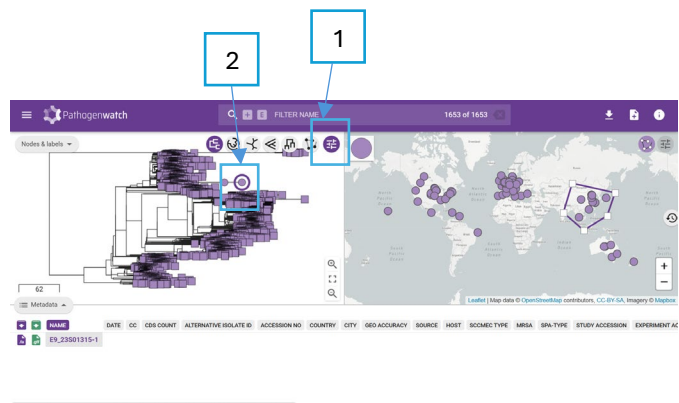
With the group selection tool (1), you can draw a line around a set of points to select. These are then highlighted in the “Tree View” and “Metadata Table”.



1. Click on this icon to see all different trees, zoom function etc.
2. You can click on leaves and nodes and they are selected with a purple halo. Selected assemblies are highlighted in the “Map View” also with a purple halo and shown exclusively in the “Metadata Tables”.

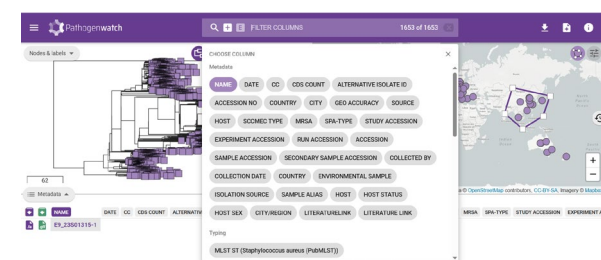
Tip:

If you right click on the tree, you can export the tree, labels etc.



The “Filter Bar” allows you to select assemblies according to their attributes in the “Metadata Tables”, including by typing assignments or antimicrobial resistance.

You can filter on multiple columns by clicking on the '+' symbol.



Download tree and files

1. In the upper right corner, you will find the download button, where you can download metadata tables, profiles and tree files.
2. To download the fasta and/or gff files, select the files in the metadata window.

